SIMATS ENGINEERING

ASSESSMENT TOOL 3 TECHNICAL PRESENTATION

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:100**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I V.T.S.CHANIKYA(192424290) certify that this submission is my original work
and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.T.S Chanikya

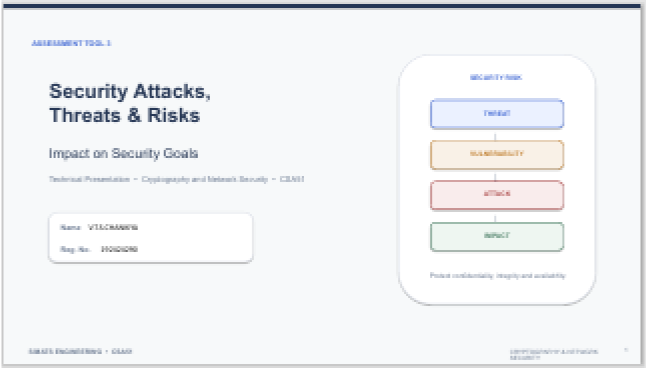
Annexure A

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Security Attacks, Threats & Risks

Impact on Security Goals

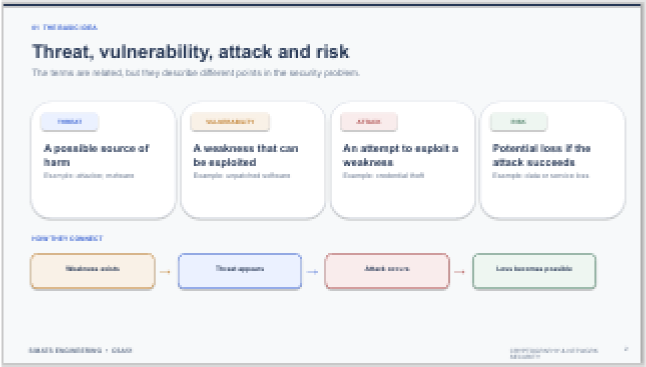
Technical Presentation · Cryptography and Network Security

Name:

Reg. No.

THREAT
VULNERABILITY
ATTACK
IMPACT

Protect confidentiality, integrity and availability



Threat, vulnerability, attack and risk

The terms are related, but they describe different points in the security problem.

A possible source of harm
Example: attacker, malware

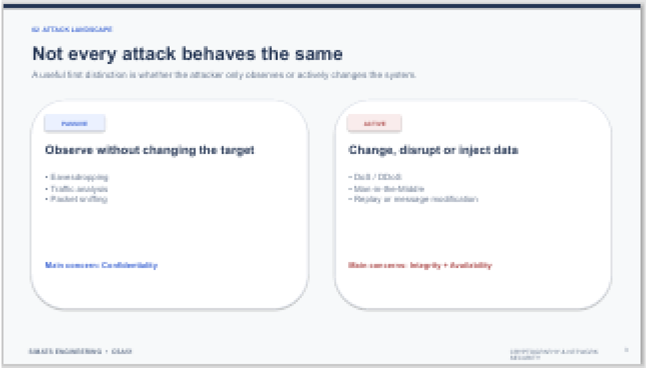
A weakness that can be exploited
Example: unpatched software

An attempt to exploit a weakness
Example: credential theft

Potential loss if the attack succeeds
Example: data or service loss

HOW THEY CONNECT

Weakness exists → Threat appears → Attack occurs → Loss becomes possible



Not every attack behaves the same

A useful first distinction is whether the attacker only observes or actively changes the system.

Observe without changing the target

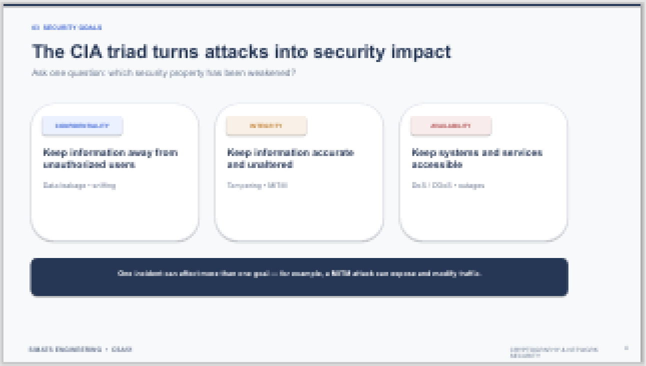
- Eavesdropping
- Traffic analysis
- Packet sniffing

Main concern: Confidentiality

Change, disrupt or inject data

- DoS / DDoS
- Man-in-the-Middle
- Replay or message modification

Main concerns: Integrity + Availability



The CIA triad turns attacks into security impact

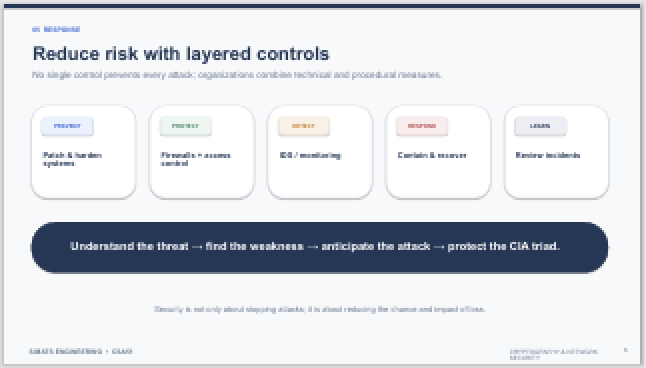
Ask one question: which security property has been weakened?

Keep information away from unauthorized users
Data leakage • sniffing

Keep information accurate and unaltered
Tampering • MITM

Keep systems and services accessible
DoS / DDoS • outages

One incident can affect more than one goal — for example, a MITM attack can expose and modify traffic.



Reduce risk with layered controls

No single control prevents every attack; organizations combine technical and procedural measures.

Patch & harden systems

Firewalls + access control

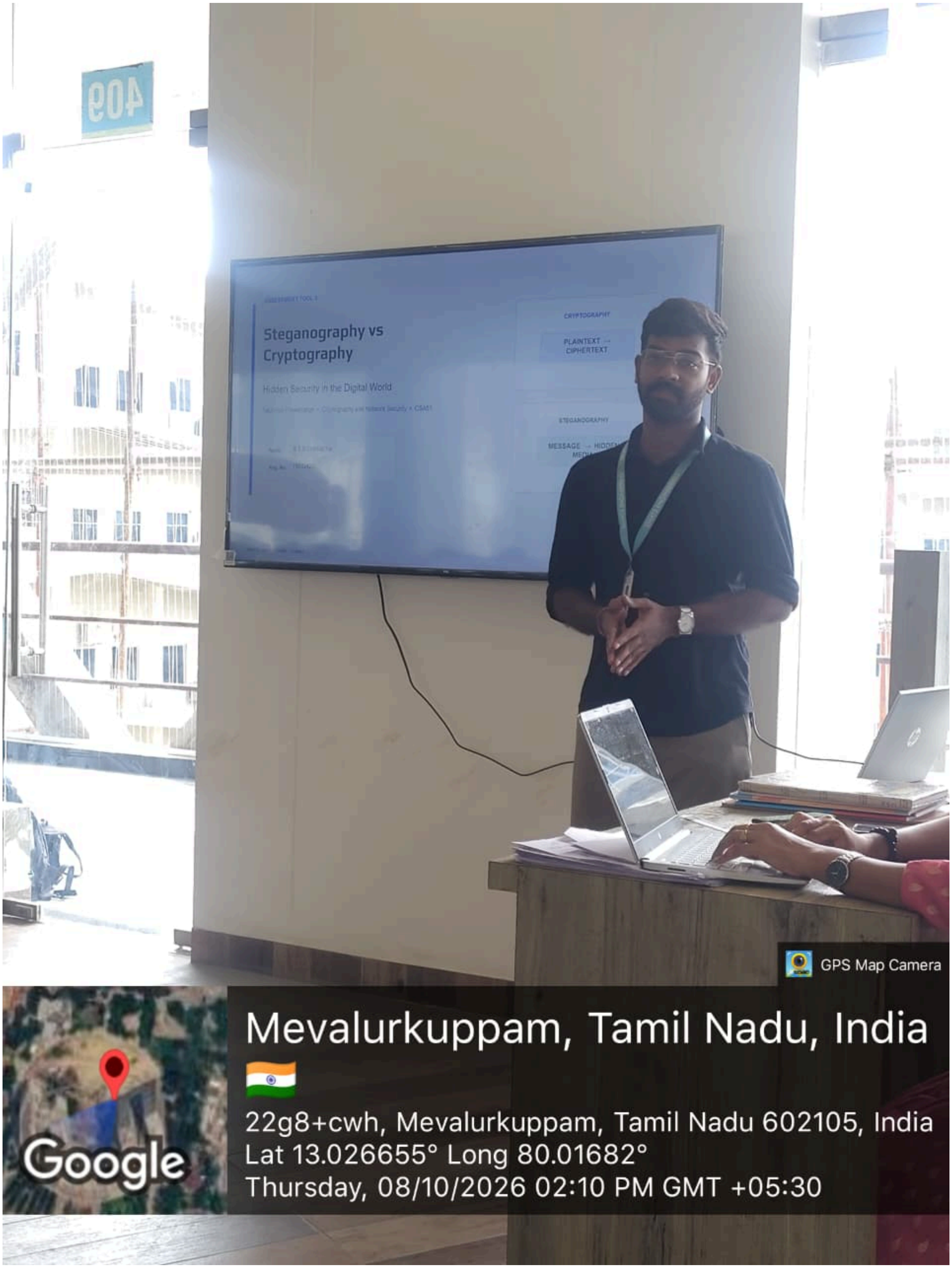
IDS / monitoring

Contain & recover

Review incidents

Understand the threat → find the weakness → anticipate the attack → protect the CIA triad.

Security is not only about stopping attacks; it is about reducing the chance and impact of loss.



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